# ECE 332 — Homework 2: Generated Answers from Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

## Problem 1 — Plane Wave $E$ and $H$ with Phase Constant 10 pts

**Setup:** 1 GHz wave traveling $+\hat{y}$ in lossless nonmagnetic medium with $\varepsilon_r = 9$. $\mathbf{E}$ along $\hat{x}$, peak 6 V/m, $|E| = 4$ V/m at $t = 0$, $y = 2$ cm.

**PLANE WAVE PARAMETERS (LOSSLESS)**

**Angular frequency** *$\omega$: rad/s; $f$: Hz*

$$\omega = 2\pi f \qquad f = \frac{\omega}{2\pi}$$

**Phase velocity** *m/s; nonmagnetic: $\mu_r = 1$*

$$u_p = \frac{c}{\sqrt{\varepsilon_r}} \qquad c = 3 \times 10^8 \text{ m/s}$$

**Wavenumber** $k$ *rad/m*

$$k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\varepsilon_r}}{c}$$

**Wavelength & impedance** *m & Ω*

$$\lambda = \frac{u_p}{f} = \frac{2\pi}{k} \text{ (m)} \qquad \eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} \ (\Omega),\ \eta_0 = 377\,\Omega$$

*Order: $\omega \to u_p \to k \to \lambda \to \eta$. Base: $\mathbf{E} = E_0\cos(\omega t \pm kx + \phi_0)\hat{e}$.*

**LOSSLESS PLANE WAVE — E AND H**

**General E**

$$\mathbf{E} = E_0\cos(\omega t - k\,\hat{k}\cdot\mathbf{r} + \phi_0)\,\hat{e}$$

*$\hat{k}$ = direction of travel (unit vector), e.g. $+y \Rightarrow \hat{k} = \hat{y}$.*
*$k$ = wavenumber (rad/m): how fast phase changes in space; $k = \omega\sqrt{\varepsilon_r}/c = 2\pi/\lambda$.*
*$\phi_0$ = initial phase constant (rad): shift set by given condition at known $t$, position.*
*$\hat{e}$ = polarization direction of $\mathbf{E}$ (unit vector), e.g. $E$ along $x \Rightarrow \hat{e} = \hat{x}$.*
*$\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, so $\hat{k}\cdot\mathbf{r}$ picks the axis, e.g. $\hat{k} = \hat{y} \Rightarrow \hat{k}\cdot\mathbf{r} = y$.*

**H from E** *$\hat{k}$, $\hat{E}$, $\hat{H}$ form right-hand set*

$$\mathbf{H} = \frac{1}{\eta}(\hat{k} \times \mathbf{E})$$

**Find $\phi_0$** *given $E = E_1$ at $t = 0$, position $r_0$*

$$E_0\cos(-kr_0 + \phi_0) = E_1 \ \Rightarrow\ \phi_0 = \cos^{-1}\!\left(\frac{E_1}{E_0}\right) + kr_0$$

*$\cos^{-1}$ gives $[0, \pi]$; take $\phi_0$ in that range.*

**Parameters from the cheat sheet order $\omega \to u_p \to k \to \lambda \to \eta$:**

$$\omega = 2\pi f = 2\pi \times 10^9 \text{ rad/s}, \qquad k = \frac{\omega\sqrt{\varepsilon_r}}{c} = \frac{2\pi\times 10^9\sqrt{9}}{3\times 10^8} = 20\pi \text{ rad/m}$$

**General E form** (wave in $+\hat{y}$ so argument is $\omega t - ky$):

$$\mathbf{E}(y,t) = 6\cos(\omega t - ky + \phi_0)\,\hat{x} \text{ V/m}$$

**Find $\phi_0$ from the given condition** $|E| = 4$ at $t = 0$, $y = 0.02$ m:

$$4 = 6\cos(-0.4\pi + \phi_0) \ \Rightarrow\ -0.4\pi + \phi_0 = \cos^{-1}(2/3) = 0.84 \text{ rad}$$

$$\boxed{\phi_0 = 2.1 \text{ rad}}$$

$$\boxed{\mathbf{E}(y,t) = 6\cos\big((2\pi\times 10^9)t - (20\pi)y + 2.1\big)\,\hat{x} \text{ V/m}}$$

**Magnetic field via $\mathbf{H} = \frac{1}{\eta}(\hat{k} \times \mathbf{E})$:**

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{377}{3} \approx 125.7\ \Omega, \qquad \hat{y}\times\hat{x} = -\hat{z}$$

$$\boxed{\mathbf{H}(y,t) = -\tfrac{6}{125.7}\cos\big((2\pi\times 10^9)t - (20\pi)y + 2.1\big)\,\hat{z} \text{ A/m}}$$

## Problem 2 — $f, u_p, \lambda, k, \eta$ and H 12 pts

**Setup:** $\mathbf{E} = 20\cos((6\pi\times10^9)t - kz)\,\hat{y}$ V/m, lossless nonmagnetic, $\varepsilon_r = 2.56$.

### PLANE WAVE PARAMETERS (LOSSLESS)

**Angular frequency** *$\omega$: rad/s; $f$: Hz*

$$\omega = 2\pi f \qquad f = \frac{\omega}{2\pi}$$

**Phase velocity** *m/s; nonmagnetic: $\mu_r = 1$*

$$u_p = \frac{c}{\sqrt{\varepsilon_r}} \qquad c = 3\times10^8 \text{ m/s}$$

**Wavenumber** $k$ *rad/m*

$$k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\varepsilon_r}}{c}$$

**Wavelength & impedance** *m & $\Omega$*

$$\lambda = \frac{u_p}{f} = \frac{2\pi}{k}\ \text{(m)} \qquad \eta = \frac{\eta_0}{\sqrt{\varepsilon_r}}\ (\Omega),\ \eta_0 = 377\,\Omega$$

*Order: $\omega \to u_p \to k \to \lambda \to \eta$. Base: $\mathbf{E} = E_0\cos(\omega t \pm kx + \phi_0)\hat{e}$.*

### $\hat{k}\times\hat{E}$ — FIND $\hat{H}$ DIRECTION

| $\hat{k}$ | $\hat{E}$ | $\hat{H}$ | $\hat{k}$ | $\hat{E}$ | $\hat{H}$ |
|---|---|---|---|---|---|
| $+\hat{z}$ | $\hat{x}$ | $+\hat{y}$ | $+\hat{z}$ | $\hat{y}$ | $-\hat{x}$ |
| $-\hat{z}$ | $\hat{x}$ | $-\hat{y}$ | $-\hat{z}$ | $\hat{y}$ | $+\hat{x}$ |
| $+\hat{y}$ | $\hat{x}$ | $-\hat{z}$ | $+\hat{y}$ | $\hat{z}$ | $+\hat{x}$ |
| $-\hat{y}$ | $\hat{x}$ | $+\hat{z}$ | $-\hat{y}$ | $\hat{z}$ | $-\hat{x}$ |
| $+\hat{x}$ | $\hat{y}$ | $+\hat{z}$ | $+\hat{x}$ | $\hat{z}$ | $-\hat{y}$ |
| $-\hat{x}$ | $\hat{y}$ | $-\hat{z}$ | $-\hat{x}$ | $\hat{z}$ | $+\hat{y}$ |

*Neg. $\hat{E}$: flip $\hat{H}$ sign. $\hat{k}\times\hat{k} = \mathbf{0}$; any $\hat{a}\times\hat{a} = \mathbf{0}$.*
*Cyclic: $\hat{x}\times\hat{y} = \hat{z}$, $\hat{y}\times\hat{z} = \hat{x}$, $\hat{z}\times\hat{x} = \hat{y}$. Reverse $\Rightarrow$ flip sign.*

**(a)** Apply the parameter chain in order:

$$\begin{aligned}
f &= \omega/(2\pi) = 3\times10^9 \text{ Hz}\\
u_p &= c/\sqrt{\varepsilon_r} = 3\times10^8/\sqrt{2.56} = 1.875\times10^8 \text{ m/s}\\
\lambda &= u_p/f = 6.25 \text{ cm}\\
k &= 2\pi/\lambda = 32\pi \approx 100.5 \text{ rad/m}\\
\eta &= \eta_0/\sqrt{\varepsilon_r} = 377/\sqrt{2.56} = 235.6\ \Omega
\end{aligned}$$

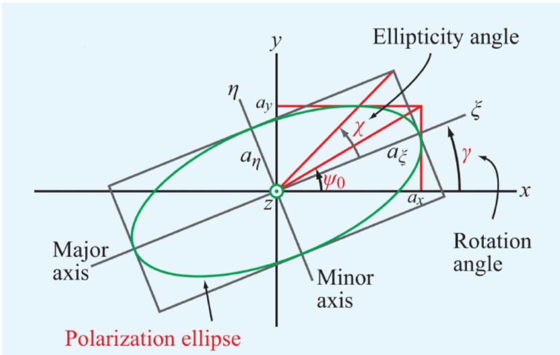
**(b) H direction from the $\hat{k}\times\hat{E}$ table:**
Wave in $+\hat{z}$, **E** along $\hat{y}$, so $\hat{z}\times\hat{y} = -\hat{x}$.

$$\boxed{\mathbf{H}(z,t) = -\tfrac{20}{235.6}\cos\big((6\pi\times10^9)t - (32\pi)z\big)\,\hat{x} = -0.0849\cos(\cdot)\,\hat{x}\ \text{A/m}}$$

## Problem 3 — Polarization Analysis (4 cases) 16 pts

**Setup:** $\mathbf{E}(z,t) = a_x\cos(\omega t - kz)\hat{x} + a_y\cos(\omega t - kz + \delta)\hat{y}$.

### POLARIZATION — GENERAL WAVE




$\mathbf{E} = a_x\cos(\omega t - kz)\hat{x} + a_y\cos(\omega t - kz + \delta)\hat{y}$

$a_x, a_y$: *bounding box half-widths (V/m); $\delta$: phase diff (rad)*

$\delta = 0$: $\mathbf{E} = (a_x\hat{x} + a_y\hat{y})\cos(\omega t - kz)$ *(linear)*

$\delta = \pi$: $\mathbf{E} = (a_x\hat{x} - a_y\hat{y})\cos(\omega t - kz)$ *(linear)*

**Auxiliary angle** $\psi_0 = \arctan(a_y/a_x) \in (0, \pi/2)$

**Rotation angle** $\gamma$ **(major axis from** $\hat{x}$**)** $\gamma \in [0, \pi)$

$$\tan(2\gamma) = \tan(2\psi_0)\cos\delta$$

*If sign($\gamma$) ≠sign($\delta$): $\gamma \leftarrow \gamma \pm \frac{\pi}{2}$.*

**Ellipticity angle** $\chi$ $\chi \in [-\pi/4, \pi/4]$

$$\sin(2\chi) = \sin(2\psi_0)\sin\delta$$

**Axial ratio** $R = 1/\tan|\chi|$ *major axis = $R\times$ minor; $R \geq 1$*

### QUADRANT RULE FOR $\tan(2\gamma)$

$N = \sin(2\psi_0)\cos\delta$, $D = \cos(2\psi_0)$:

| $N$ | $D$ | $2\gamma$ (deg) | $2\gamma$ (rad) |
|---|---|---|---|
| + | + | $\arctan(N/D)$ | $\arctan(N/D)$ |
| + | − | $180^\circ - \arctan\lvert N/D\rvert$ | $\pi - \arctan\lvert N/D\rvert$ |
| − | − | $180^\circ + \arctan\lvert N/D\rvert$ | $\pi + \arctan\lvert N/D\rvert$ |
| − | + | $360^\circ - \arctan\lvert N/D\rvert$ | $2\pi - \arctan\lvert N/D\rvert$ |

$\gamma = (2\gamma)/2$. $D = 0 \Rightarrow 2\gamma = \frac{\pi}{2} \Rightarrow \gamma = \frac{\pi}{4}$.

*Convert:* $\theta_{rad} = \theta_\circ \times \frac{\pi}{180}$; $\theta_\circ = \theta_{rad} \times \frac{180}{\pi}$.

### $\chi$ SHAPE & HANDEDNESS

*$\chi$ describes the shape of the ellipse and rotation direction.*

| $\chi$ (rad) | Polarization Type | Handedness |
|---|---|---|
| 0 | Linear | None |
| $(0, \pi/4)$ | RHEP | RH (CCW) |
| $\pi/4$ | RHC | RH (CCW) |
| $-\pi/4$ | LHC | LH (CW) |
| $(-\pi/4, 0)$ | LHEP | LH (CW) |

### CIRCULAR POLARIZATION — TIME DOMAIN

*$a_x = a_y = a$, wave in $+\hat{z}$*

**RHC** $\delta = -\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} - j\hat{y})e^{-jkz}$

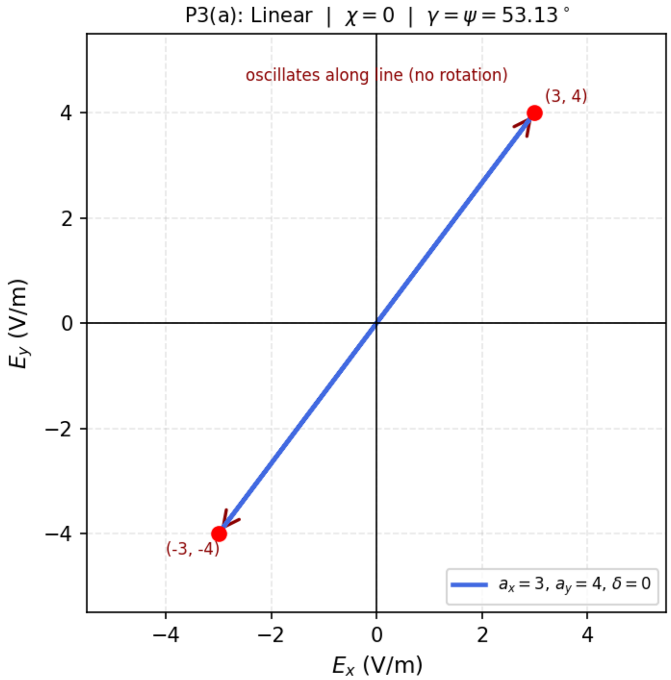
$$\mathbf{E}(z,t) = a\left[\cos(\omega t - kz)\,\hat{x} + \sin(\omega t - kz)\,\hat{y}\right]$$

**LHC** $\delta = +\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} + j\hat{y})e^{-jkz}$

$$\mathbf{E}(z,t) = a\left[\cos(\omega t - kz)\,\hat{x} - \sin(\omega t - kz)\,\hat{y}\right]$$

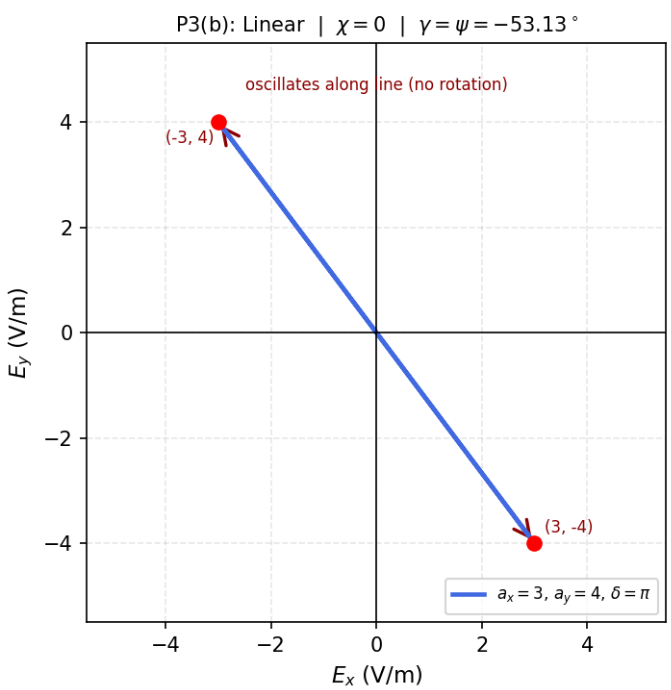
**(a)** $a_x = 3$, $a_y = 4$, $\delta = 0$: $\delta = 0 \Rightarrow$ linear. $R = \infty$, $\chi = 0$. Rotation angle = inclination:

$$\psi = \tan^{-1}(4/3) = 0.9273 \text{ rad} = 53.13^\circ \Rightarrow \boxed{\gamma = 53.13^\circ,\ \chi = 0}$$




**(b)** $a_x = 3$**,** $a_y = 4$**,** $\delta = \pi$**:** $\delta = \pi \Rightarrow$ linear again ($\cos(\theta+\pi) = -\cos\theta$, so $\mathbf{E} = (3\hat{x}-4\hat{y})\cos(\omega t-kz)$). $\chi = 0$.

$$\psi = \tan^{-1}(-4/3) = -0.9273 \text{ rad} = -53.13^\circ \Rightarrow \boxed{\gamma = -53.13^\circ,\ \chi = 0}$$



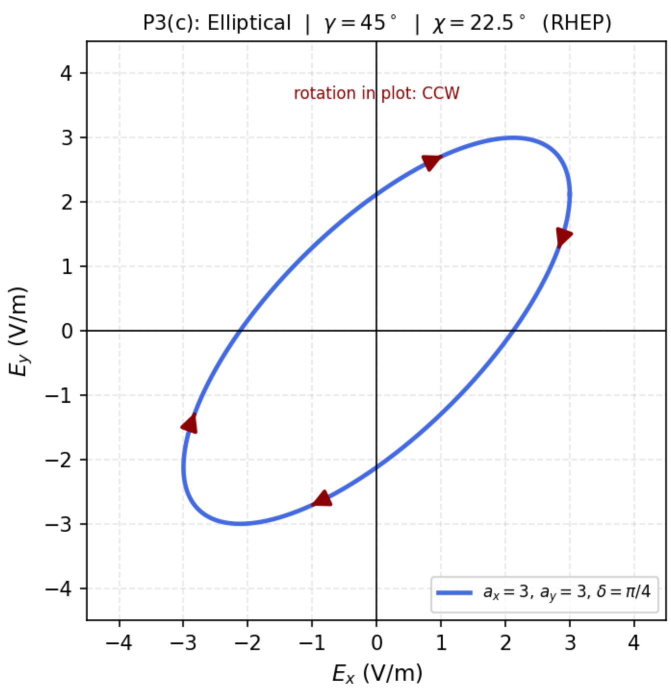
**(c) $a_x = 3$, $a_y = 3$, $\delta = \pi/4$:** elliptical. Auxiliary $\psi_0 = \tan^{-1}(1) = \pi/4$.

$$\tan 2\gamma = \tan(\pi/2)\cos(\pi/4) = \infty\sqrt{2}/2 \Rightarrow 2\gamma = \pi/2$$

$$\sin 2\chi = \sin(\pi/2)\sin(\pi/4) = \sqrt{2}/2 \Rightarrow 2\chi = \pi/4$$

$$\boxed{\gamma = \pi/4 = 45^\circ, \quad \chi = \pi/8 = 22.5^\circ}$$

$\chi > 0 \Rightarrow$ right-hand elliptical (RHEP).




**(d) $a_x = 3$, $a_y = 4$, $\delta = -3\pi/4$:** elliptical. $\psi_0 = \tan^{-1}(4/3) \approx 0.9273$, $2\psi_0 \approx 1.8546$ (106.26°).

$$\cos\delta = -\sqrt{2}/2, \quad \sin\delta = -\sqrt{2}/2$$

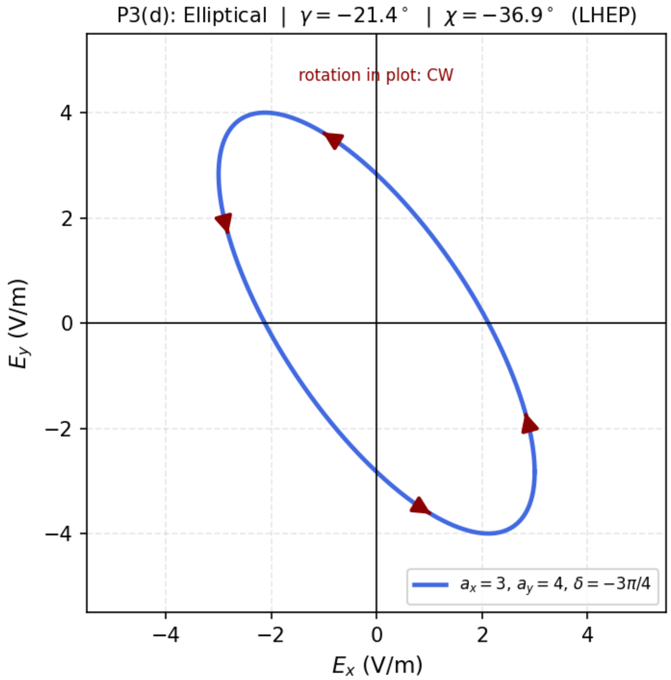
$$\tan 2\gamma = \tan(1.8546)\cdot(-\sqrt{2}/2) \approx 2.4244$$

$N = \sin(2\psi_0)\cos\delta < 0$, $D = \cos(2\psi_0) < 0$ (quadrant rule: both negative)

$$2\gamma \approx -1.1796 \text{ rad} = -73.73^\circ \Rightarrow \boxed{\gamma \approx -21.4^\circ}$$

$$\sin 2\chi = \sin(1.8546)\cdot(-\sqrt{2}/2) = -0.7462 \Rightarrow 2\chi \approx -0.7462 \text{ rad} \Rightarrow \boxed{\chi \approx -36.9^\circ}$$

$\chi < 0 \Rightarrow$ left-hand elliptical (LHEP).


P3(d): Elliptical | $\gamma = -21.4^\circ$ | $\chi = -36.9^\circ$ (LHEP)
rotation in plot: CW
$E_y$ (V/m)
$E_x$ (V/m)
$a_x = 3$, $a_y = 4$, $\delta = -3\pi/4$

## Problem 4 — Linear = RHC + LHC Decomposition 8 pts

**Setup:** Show $\tilde{\mathbf{E}} = a_x e^{-jkz}\hat{x}$ can be written as $\tilde{\mathbf{E}}_R + \tilde{\mathbf{E}}_L$.

### CIRCULAR POLARIZATION — TIME DOMAIN

$a_x = a_y = a$, wave in $+\hat{z}$

**RHC** $\delta = -\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} - j\hat{y})e^{-jkz}$

$$\mathbf{E}(z,t) = a\left[\cos(\omega t - kz)\,\hat{x} + \sin(\omega t - kz)\,\hat{y}\right]$$

**LHC** $\delta = +\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} + j\hat{y})e^{-jkz}$

$$\mathbf{E}(z,t) = a\left[\cos(\omega t - kz)\,\hat{x} - \sin(\omega t - kz)\,\hat{y}\right]$$

**From the cheat sheet definitions:**

$$\tilde{\mathbf{E}}_R = a_R(\hat{x} - j\hat{y})e^{-jkz}, \qquad \tilde{\mathbf{E}}_L = a_L(\hat{x} + j\hat{y})e^{-jkz}$$

**Try $a_R = a_L = a_x/2$:**

$$\begin{aligned}\tilde{\mathbf{E}}_R + \tilde{\mathbf{E}}_L &= \left[(a_R + a_L)\hat{x} + j(-a_R + a_L)\hat{y}\right]e^{-jkz} \\ &= \left[a_x\,\hat{x} + j\cdot 0\cdot\hat{y}\right]e^{-jkz} = a_x e^{-jkz}\hat{x} = \tilde{\mathbf{E}}\ \checkmark\end{aligned}$$

$$\boxed{a_R = a_L = a_x/2}$$

## Problem 5 — Dry Soil at 4 Frequencies (Loss Tangent Classification) 24 pts

**Setup:** $\varepsilon_r = 2.5$, $\mu_r = 1$, $\sigma = 10^{-4}$ S/m.

### LOSSY MEDIA — CLASSIFICATION

Loss tangent

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0}$$

| $\varepsilon''/\varepsilon'$ | Type | Use |
|---|---|---|
| $> 100$ | Good conductor | GC approx |
| 0.01 to 100 | Quasi-conductor | Exact eqs |
| $< 0.01$ | Low-loss dielectric | LD approx |

*$\varepsilon' = \varepsilon_r\varepsilon_0$; $\varepsilon'' = \sigma/\omega$; nonmag.: $\mu = \mu_0$; free space: $\varepsilon_r = \mu_r = 1$.*

### ANY MEDIUM — EXACT ($X = \sqrt{1 + (\varepsilon''/\varepsilon')^2}$)

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X-1}\ \text{Np/m}, \quad \beta = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X+1}\ \text{rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{u_p}{f}\ \text{(m)}, \quad u_p = \frac{\omega}{\beta}\ \text{(m/s)}$$

$$\eta_c = \frac{\eta}{X^{1/2}}e^{j\theta_\eta}\ (\Omega), \quad \theta_\eta = \tfrac{1}{2}\arctan\frac{\sigma}{\omega\varepsilon'}\ \text{(rad)}$$

*Rect:* $\eta_c = |\eta_c|\cos\theta_\eta + j|\eta_c|\sin\theta_\eta$, $|\eta_c| = \eta/X^{1/2}$ $\eta = \eta_0/\sqrt{\varepsilon_r}$; $\varepsilon' = \varepsilon_r\varepsilon_0$.
*Given* $\tilde{\mathbf{H}} \propto e^{-\alpha y}e^{-j\beta y}$: $\gamma = \alpha + j\beta$; $\eta_c = j\omega\mu_0/\gamma = \omega\mu_0(\beta + j\alpha)/(\alpha^2 + \beta^2)$

### LOW-LOSS MEDIUM ($\varepsilon''/\varepsilon' \ll 1$)

$$\alpha \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon'}} = \frac{\sigma\eta}{2}\ \text{(Np/m)}$$

$$\beta \approx \omega\sqrt{\mu\varepsilon'} = \frac{\omega\sqrt{\varepsilon_r}}{c}\ \text{(rad/m)}, \quad u_p \approx \frac{c}{\sqrt{\varepsilon_r}}\ \text{(m/s)}$$

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon'}} = \eta\ (\Omega,\ \angle \approx 0)$$

### GOOD CONDUCTOR ($\varepsilon''/\varepsilon' \gg 1$)

$$\alpha \approx \beta \approx \sqrt{\pi f\mu\sigma}\quad \text{(Np/m, rad/m)}$$

$$u_p \approx \sqrt{4\pi f/\mu\sigma}\ \text{(m/s)}, \quad \lambda \approx u_p/f\ \text{(m)}$$

$$\eta_c \approx (1+j)\sqrt{\frac{\pi f\mu}{\sigma}} \approx (1+j)\frac{\alpha}{\sigma}\quad (\Omega)$$

**Compute the loss tangent ratio:**

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0} = \frac{\sigma/(2\pi\varepsilon_r\varepsilon_0)}{f} = \frac{7.193\times 10^5\ \text{Hz}}{f}$$

| $f$ | $\varepsilon''/\varepsilon'$ | Class | Use |
|---|---|---|---|
| 60 Hz | $1.20 \times 10^4$ | Good conductor | $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$ |
| 1 kHz | 719.3 | Good conductor | GC approx |
| 1 MHz | 0.719 | Quasi-conductor | Exact eqs |
| 1 GHz | $7.19 \times 10^{-4}$ | Low-loss | LD approx |

**(a) 60 Hz** (good conductor):

$$\alpha \approx \beta = \sqrt{\pi(60)(4\pi \times 10^{-7})(10^{-4})} = 4.87 \times 10^{-6} \text{ Np/m, rad/m}$$

$$\lambda = 2\pi/\beta = 1.29 \times 10^6 \text{ m}, \quad u_p = \sqrt{4\pi f/\mu\sigma} = 7.74 \times 10^7 \text{ m/s}$$

$$\eta_c \approx (1+j)\alpha/\sigma = (1+j)(0.0487)\ \Omega$$

**(b) 1 kHz** (good conductor): same formulas, scale by $\sqrt{1000/60} \approx 4.08\times$:

$$\alpha \approx \beta = 1.99 \times 10^{-5} \text{ Np/m}, \quad \lambda \approx 3.16 \times 10^5 \text{ m}$$

**(c) 1 MHz** (quasi-conductor): use exact $X = \sqrt{1 + (\varepsilon''/\varepsilon')^2} = \sqrt{1 + 0.719^2} = 1.231$:

$$\alpha = \omega\sqrt{\tfrac{\mu\varepsilon'}{2}}\sqrt{X-1} \text{ Np/m}, \quad \beta = \omega\sqrt{\tfrac{\mu\varepsilon'}{2}}\sqrt{X+1} \text{ rad/m}$$

$$\eta_c = \tfrac{\eta}{\sqrt{X}} e^{j\theta_\eta}, \quad \theta_\eta = \tfrac{1}{2}\arctan(0.719) = 17.7^\circ$$

**(d) 1 GHz** (low-loss dielectric):

$$\alpha \approx \tfrac{\sigma}{2}\sqrt{\mu/\varepsilon'} = \tfrac{10^{-4}}{2} \cdot \tfrac{377}{\sqrt{2.5}} = 0.0119 \text{ Np/m}$$

$$\beta \approx \omega\sqrt{\mu\varepsilon'} = \tfrac{2\pi \times 10^9 \sqrt{2.5}}{3 \times 10^8} = 33.1 \text{ rad/m}$$

$$\lambda \approx 2\pi/\beta = 0.190 \text{ m}, \ u_p \approx c/\sqrt{\varepsilon_r} = 1.90 \times 10^8 \text{ m/s}, \ \eta_c \approx 238.4\ \Omega$$

## Problem 6 — Phasor → Time Domain (Lossy) 10 pts

**Setup:** 300 MHz, nonmagnetic lossy, $\tilde{\mathbf{H}} = (\hat{x} - j4\hat{z})e^{-2y}e^{-j9y}$ A/m.

**PHASOR ↔ TIME DOMAIN**

| Phasor coeff | $\mathrm{Re}\{(\cdot)\, e^{j\omega t}e^{-j\beta z}\}$ | Result |
|---|---|---|
| $A$ | $\mathrm{Re}\{Ae^{j(\omega t-\beta z)}\}$ | $A\cos(\omega t-\beta z)$ |
| $-jA$ | $\mathrm{Re}\{Ae^{j(\omega t-\beta z-\pi/2)}\}$ | $A\sin(\omega t-\beta z)$ |
| $+jA$ | $\mathrm{Re}\{Ae^{j(\omega t-\beta z+\pi/2)}\}$ | $-A\sin(\omega t-\beta z)$ |
| $-A$ | $\mathrm{Re}\{Ae^{j(\omega t-\beta z\pm\pi)}\}$ | $-A\cos(\omega t-\beta z)$ |
| $Ae^{j\phi}$ | $\mathrm{Re}\{Ae^{j(\omega t-\beta z+\phi)}\}$ | $A\cos(\omega t-\beta z+\phi)$ |
| $A+jB$ | $\mathrm{Re}\{\lvert Z\rvert e^{j(\omega t-\beta z+\theta)}\}$ | $\lvert Z\rvert\cos(\omega t-\beta z+\theta)$ |

$|Z| = \sqrt{A^2+B^2}$, $\theta = \arctan(B/A)$ (check quadrant)

**$\hat{k}\times\hat{E}$ — FIND $\hat{H}$ DIRECTION**

| $\hat{k}$ | $\hat{E}$ | $\hat{H}$ | $\hat{k}$ | $\hat{E}$ | $\hat{H}$ |
|---|---|---|---|---|---|
| $+\hat{z}$ | $\hat{x}$ | $+\hat{y}$ | $+\hat{z}$ | $\hat{y}$ | $-\hat{x}$ |
| $-\hat{z}$ | $\hat{x}$ | $-\hat{y}$ | $-\hat{z}$ | $\hat{y}$ | $+\hat{x}$ |
| $+\hat{y}$ | $\hat{x}$ | $-\hat{z}$ | $+\hat{y}$ | $\hat{z}$ | $+\hat{x}$ |
| $-\hat{y}$ | $\hat{x}$ | $+\hat{z}$ | $-\hat{y}$ | $\hat{z}$ | $-\hat{x}$ |
| $+\hat{x}$ | $\hat{y}$ | $+\hat{z}$ | $+\hat{x}$ | $\hat{z}$ | $-\hat{y}$ |
| $-\hat{x}$ | $\hat{y}$ | $-\hat{z}$ | $-\hat{x}$ | $\hat{z}$ | $+\hat{y}$ |

*Neg. $\hat{E}$: flip $\hat{H}$ sign. $\hat{k}\times\hat{k}=\mathbf{0}$; any $\hat{a}\times\hat{a}=\mathbf{0}$.*
*Cyclic: $\hat{x}\times\hat{y}=\hat{z}$, $\hat{y}\times\hat{z}=\hat{x}$, $\hat{z}\times\hat{x}=\hat{y}$. Reverse ⇒ flip sign.*

**Read off:** $\omega = 6\pi\times10^8$ rad/s, $\alpha = 2$ Np/m, $\beta = 9$ rad/m, $\hat{k} = \hat{y}$.

**Time-domain H** using the phasor table ($-j \to \sin$):

$$\mathbf{H}(y,t) = \left[\cos(\omega t - 9y)\,\hat{x} \;-\; 4\cos(\omega t - 9y + \tfrac{\pi}{2})\,\hat{z}\right]e^{-2y} \text{ A/m}$$

**Find $\tilde{\mathbf{E}}$** using $\tilde{\mathbf{E}} = -\eta_c(\hat{k}\times\tilde{\mathbf{H}})$, but for lossy we need $\eta_c$ first. Compute loss tangent:

$$\varepsilon''/\varepsilon' = (\alpha\beta\cdot 2)/(\omega^2\mu\varepsilon') \quad \text{(implicit from } \alpha, \beta)$$

Or compute $\eta_c = j\omega\mu_0/\gamma$ with $\gamma = \alpha + j\beta = 2 + j9$:

$$\eta_c = \frac{j(6\pi\times10^8)(4\pi\times10^{-7})}{2+j9} = \frac{j(753.98)}{2+j9} = \frac{j753.98\,(2-j9)}{4+81} = \frac{6785.8+j1507.9}{85} \approx 79.8 + j17.7\ \Omega$$

**Then $\hat{y}\times\hat{x} = -\hat{z}$, $\hat{y}\times\hat{z} = \hat{x}$:**

$$\tilde{\mathbf{E}} = -\eta_c\big(\hat{y}\times(\hat{x} - j4\hat{z})\big)e^{-2y}e^{-j9y} = -\eta_c(-\hat{z} - j4\hat{x})e^{-2y}e^{-j9y}$$

$$\tilde{\mathbf{E}} = \eta_c(j4\hat{x} + \hat{z})e^{-2y}e^{-j9y} \text{ V/m}$$

**Time domain:**

$$\mathbf{E}(y,t) = |\eta_c|\left[4\cos(\omega t - 9y + \pi/2 + \theta_\eta)\,\hat{x} + \cos(\omega t - 9y + \theta_\eta)\,\hat{z}\right]e^{-2y}$$

where $|\eta_c| \approx 81.8$, $\theta_\eta = \arctan(17.7/79.8) \approx 12.5°$.

## Problem 7 — Direction of Travel and Poynting 10 pts

**Setup:** $\mathbf{E} = 3\cos(\pi\times10^7 t + kx)\hat{y} - 2\cos(\pi\times10^7 t + kx)\hat{z}$ V/m, nonmagnetic $\varepsilon_r = 9$.

**DIRECTION OF PROPAGATION**

| Arg. contains | $\hat{k}$ | Arg. contains | $\hat{k}$ |
|---|---|---|---|
| $\omega t - kz$ | $+\hat{z}$ | $\omega t + kz$ | $-\hat{z}$ |
| $\omega t - ky$ | $+\hat{y}$ | $\omega t + ky$ | $-\hat{y}$ |
| $\omega t - kx$ | $+\hat{x}$ | $\omega t + kx$ | $-\hat{x}$ |

*Minus before $k$-axis $\Rightarrow$ + direction. Plus $\Rightarrow$ − direction.*

**AVERAGE POWER DENSITY (POYNTING)**

**General** $W/m^2$

$$\mathbf{S}_{\text{av}} = \frac{1}{2}\,\text{Re}\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\}$$

**Lossless shortcut** $|E_0|^2 = a_x^2 + a_y^2 + a_z^2$

$$\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta}\,\hat{k}\ (\text{W/m}^2)$$

**Lossy medium** $\eta_c = |\eta_c|e^{j\theta_\eta}$; *wave in* $+\hat{z}$

$$\mathbf{S}_{\text{av}} = \frac{|\tilde{\mathbf{E}}(0)|^2}{2|\eta_c|}e^{-2\alpha z}\cos\theta_\eta\ \hat{z}\ (\text{W/m}^2)$$

*Direction = $\hat{k}$. If wave goes in $-\hat{x}$, then $\mathbf{S}_{av}$ is a negative $x$-component.*

**Direction from the table:** argument is $\omega t + kx$ (plus sign $\Rightarrow$ wave travels in $-\hat{x}$).

$$\boxed{\hat{k} = -\hat{x}}$$

No $e^{-\alpha x}$ attenuation factor, so the medium is effectively lossless here.

**Intrinsic impedance:**

$$\eta = \eta_0/\sqrt{\varepsilon_r} = 377/3 \approx 125.7\ \Omega = 40\pi\ \Omega$$

**Poynting (lossless shortcut):** $|E_0|^2 = 3^2 + 2^2 = 13$.

$$\mathbf{S}_{\text{av}} = \tfrac{|E_0|^2}{2\eta}\hat{k} = \tfrac{13}{80\pi}(-\hat{x})\ \text{W/m}^2 \approx -51.7\ \text{mW/m}^2\,\hat{x}$$

$$\boxed{\mathbf{S}_{\text{av}} \approx -51.7\ \text{mW/m}^2\ \hat{x}}$$